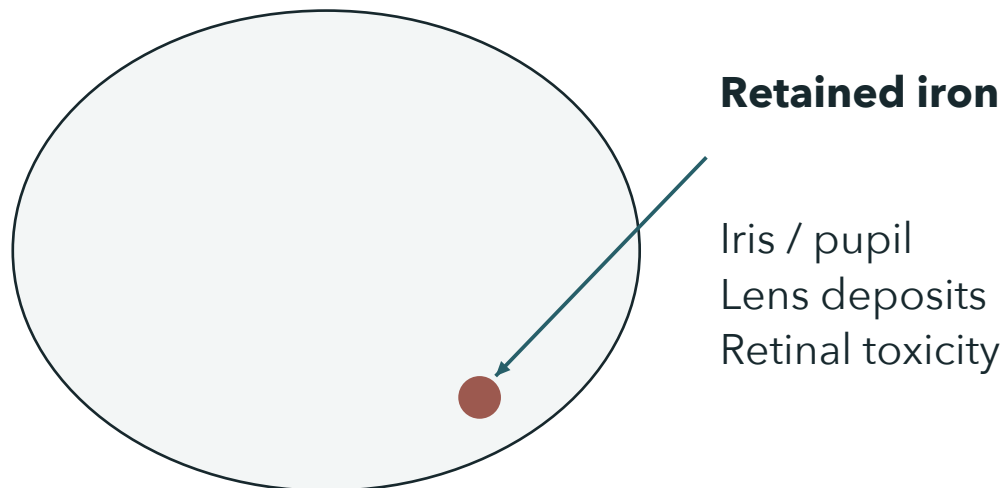


Late posterior complications after ocular trauma

Part 2 | Retained iron: toxicity can outlast the wound

An apparently healed entry site does not exclude siderosis.

ERG can reveal functional injury before severe acuity loss.



Original causal sketch: retained iron can affect lens, pupil and retinal function.

Revisit high-velocity metal exposure when pigmentation, cataract, pupillary change or unexplained retinal dysfunction develops. Localise a possible foreign body using an appropriate imaging route before considering removal. Electrophysiology can detect functional involvement, but recovery depends on established damage. Explain why a quiet anterior segment is insufficient reassurance after a plausible penetrating mechanism.

A recent traumatic macular hole

Selected small holes may close without immediate surgery.

Recent small hole

Possible closure



Serial OCT documents anatomy. Closure does not guarantee normal photoreceptor function.

Original OCT sequence: a small recent full-thickness defect can close.

Consider age, minimum defect size, vitreous state, time since injury and associated macular damage. Observation with serial OCT can be reasonable in a selected young patient, but it needs a defined review and escalation plan. Recent retrospective paediatric data support spontaneous closure in some eyes; they do not establish a universal waiting interval or guarantee useful visual recovery.

Closure and useful sight are different outcomes

Outer-retinal injury can limit vision despite anatomical recovery.

Finding	Implication
Hole closes	Assess the recovered tissue and residual function
Ellipsoid or RPE damage	Potential limit to visual recovery
Associated commotio or rupture	May explain persistent symptoms
Persistent or enlarging defect	Reconsider specialist operative options

Compare the evolving anatomy with acuity and distortion rather than treating closure as a complete outcome. The injury may have damaged photoreceptors beyond the defect. Counsel the patient about what further surgery can and cannot change. A decision to intervene should name the remaining mechanism and its realistic functional target.

Indirect traumatic optic neuropathy

Routine megadose steroid or canal surgery lacks established benefit.

Finding	Implication
Indirect TON	Assess nerve dysfunction after excluding ocular causes
Orbital compartment syndrome	Separate emergency requiring decompression
Concomitant significant head injury	Megadose steroids can increase mortality
Routine canal decompression	No consistent proven benefit in indirect TON

Distinguish a compressive emergency or specific displaced fragment from indirect injury without a demonstrated surgical target. Stabilise the patient and discuss uncertainty with the specialist team. An intervention's biological rationale is not proof of visual benefit, and systemic harm can outweigh an unproven ocular gain. Do not use the word trauma as one treatment category.

Delayed blur: reassess the full pathway

Late change can arise in several tissues after the same injury.

Finding	Implication
Lens or pressure change	Reassess optics and anterior-segment sequelae
New field symptoms	Look for breaks, dialysis or detachment
Macular change	Review OCT and outer-retinal integrity
Afferent deficit	Assess optic-nerve function

Use the original mechanism and new symptom to choose a targeted examination. A stable wound or an attached retina does not end surveillance of other structures. Explain which change needs urgent reassessment and which can be monitored. The follow-up plan should preserve the history of trauma so that a later clinician can recognise delayed structural or functional consequences.